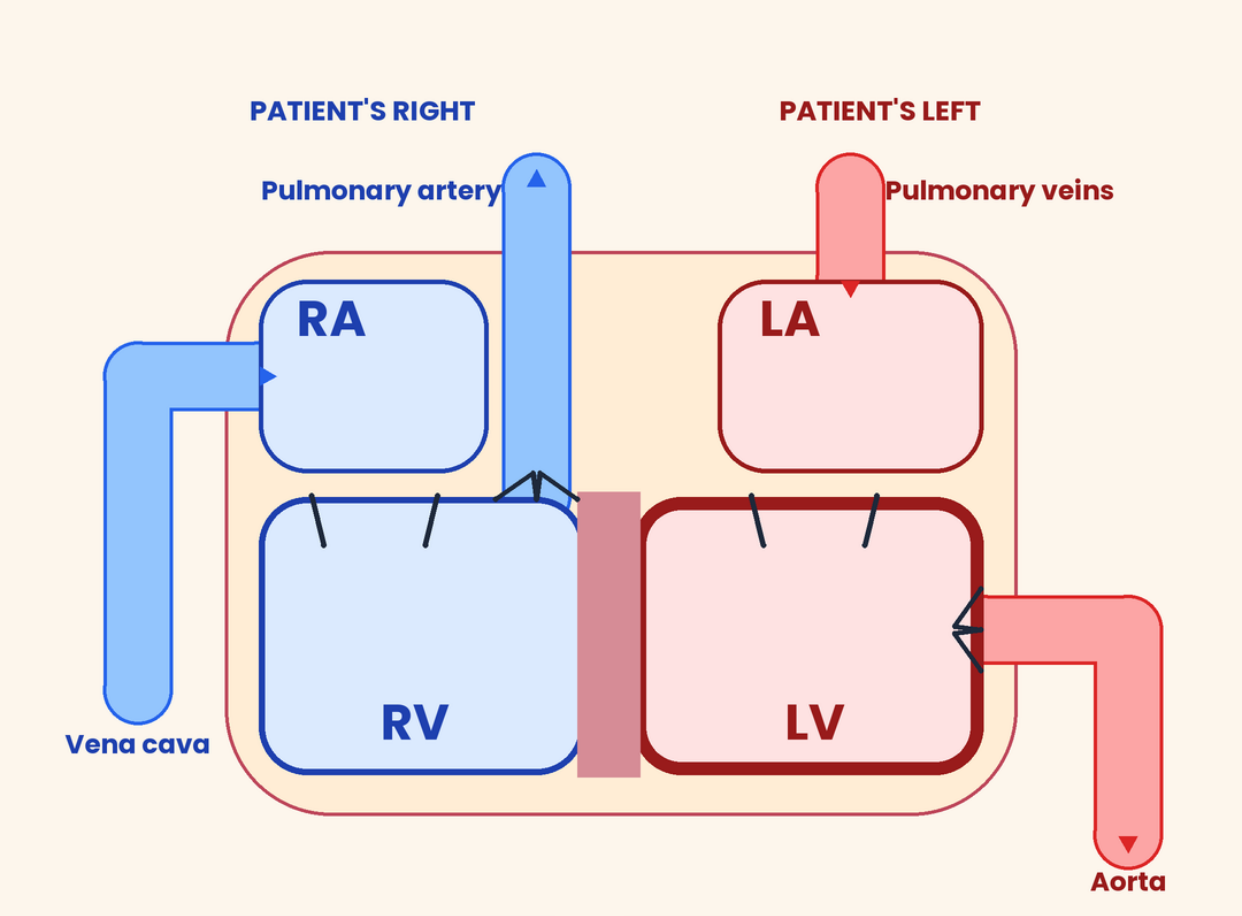
The Human Heart

The heart is a muscular pump that keeps blood moving non-stop, about 70-75 beats every minute, so every cell gets oxygen and food and gets rid of carbon dioxide and waste.



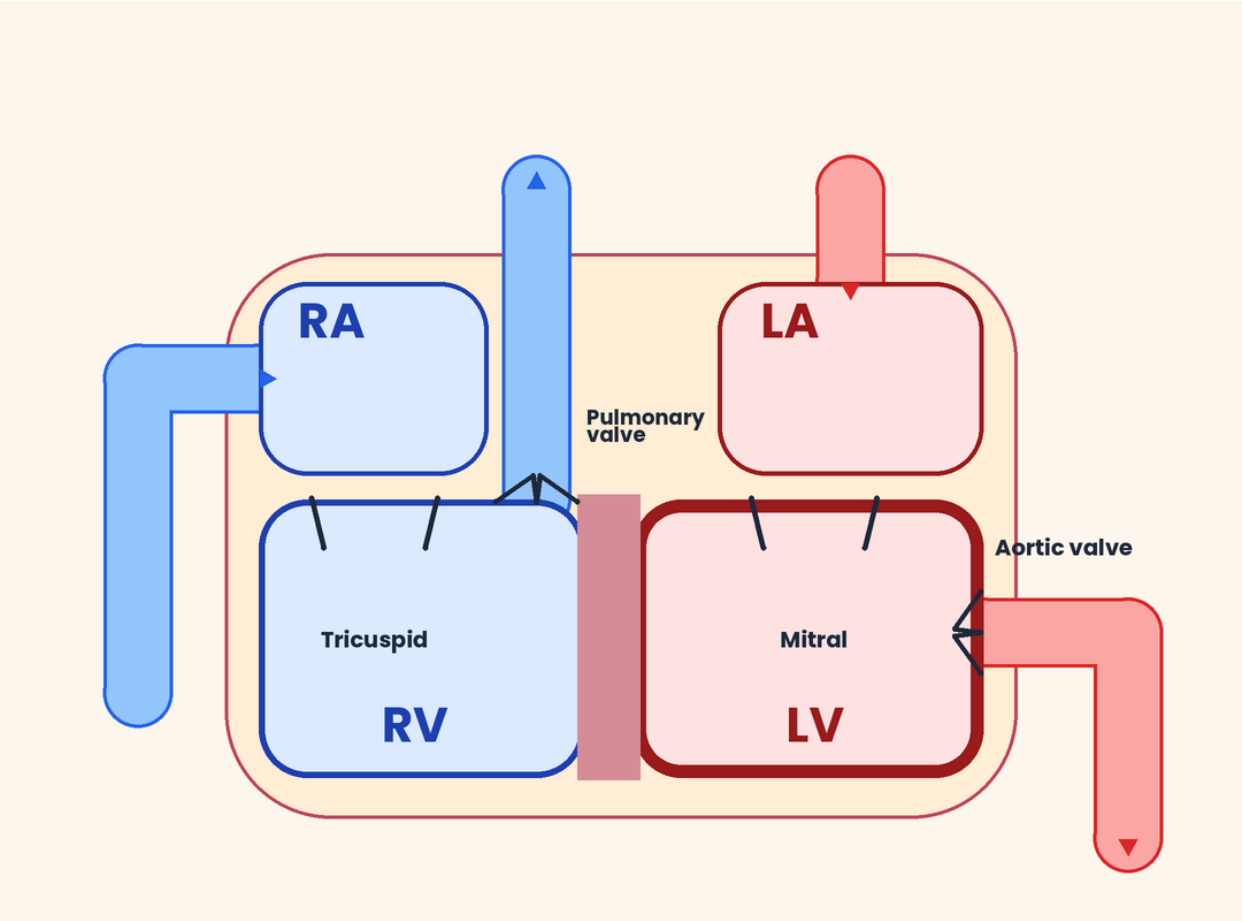
Four chambers

Atria (upper: right atrium RA, left atrium LA) receive blood coming into the heart. Ventricles (lower: right ventricle RV, left ventricle LV) pump blood out of the heart. Ventricle walls are thicker than atrial walls; the left ventricle has the thickest wall because it pumps blood to the whole body.

The septum is the muscular wall between the right and left sides, so oxygen-rich and oxygen-poor blood never mix. Right side = deoxygenated blood, left side = oxygenated blood.

3D heart model: BodyParts3D, (c) The Database Center for Life Science, licensed under CC BY-SA 2.1 Japan.

Valves = one-way doors



Valves open to let blood flow forward and snap shut to stop it flowing backward.

AV (atrioventricular) valves

Tricuspid valve (3 flaps): right atrium to right ventricle. Mitral or bicuspid valve (2 flaps): left atrium to left ventricle.

Semilunar valves (exit doors)

Pulmonary valve: right ventricle to pulmonary artery. Aortic valve: left ventricle to aorta.

Memory aid: LAB RAT

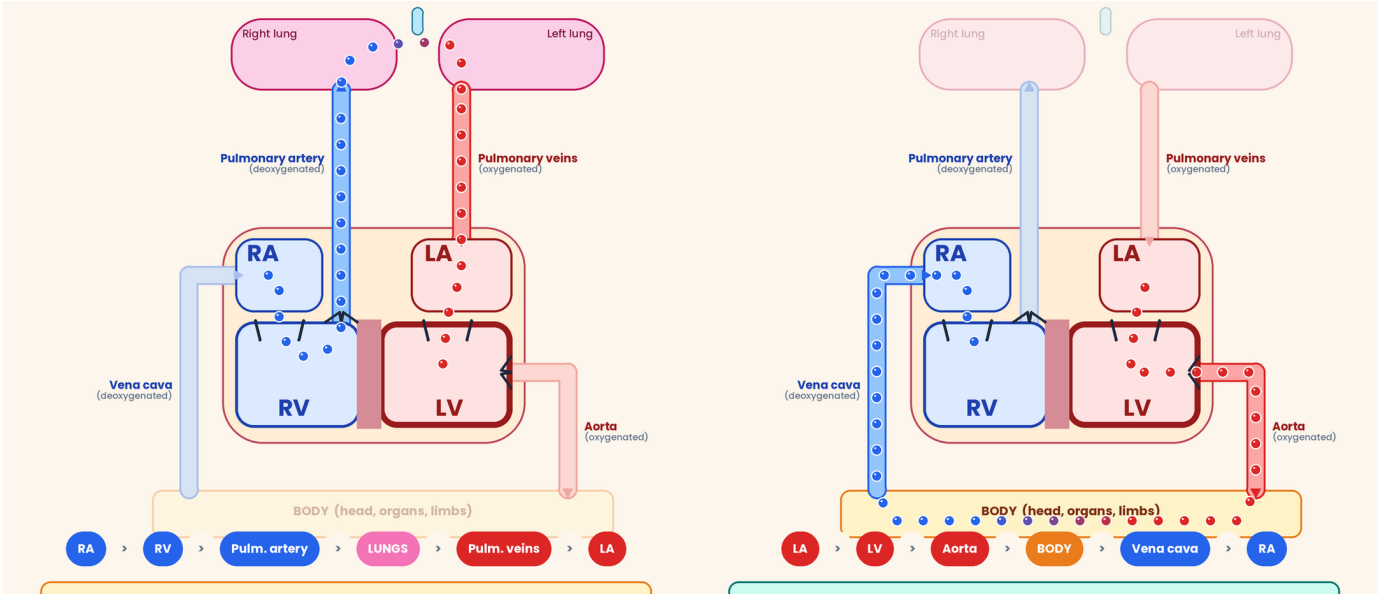
Left Atrium = Bicuspid, Right Atrium = Tricuspid.

Heart sounds: lub-dub

LUB = tricuspid and mitral valves close as the ventricles contract. DUB = aortic and pulmonary valves close as the ventricles relax.

3D heart model: BodyParts3D, (c) The Database Center for Life Science, licensed under CC BY-SA 2.1 Japan.

Double circulation



Loop 1 - Pulmonary:

Right atrium, right ventricle, pulmonary artery, lungs (blood picks up oxygen and gives out carbon dioxide), pulmonary veins, left atrium.

Loop 2 - Systemic:

Left atrium, left ventricle, aorta, whole body, vena cava, right atrium.

Exception to remember

Pulmonary artery carries DEOXYGENATED blood; pulmonary veins carry OXYGENATED blood.

Blood passes through the heart twice in one full circuit, so this is called double circulation.

3D heart model: BodyParts3D, (c) The Database Center for Life Science, licensed under CC BY-SA 2.1 Japan.

Solved example: cardiac output

A person's heart beats 72 times per minute, pumping 70 mL of blood per beat. Find the cardiac output per minute.

Given

Heart rate (HR) = 72 beats/min; Stroke volume (SV) = 70 mL per beat.

Formula

Cardiac output = HR x SV

Substitute

= 72 x 70 = 5040 mL/min

Answer

Cardiac output = 5040 mL/min, about 5 litres of blood per minute.

NEET shortcuts

1. Right side = deoxygenated blood; left side = oxygenated blood.
2. Exception: pulmonary artery = deoxygenated, pulmonary vein = oxygenated.
3. LAB RAT: Left Atrium Bicuspid, Right Atrium Tricuspid.
4. Left ventricle has the thickest wall.
5. LUB = AV valves close; DUB = semilunar (aortic and pulmonary) valves close.

Quick check

Q1. Which valve lies between the left atrium and left ventricle? Ans: Mitral (bicuspid) valve.

Q2. Which vessel carries oxygenated blood to the left atrium? Ans: Pulmonary vein.

Q3. Heart rate 80/min, stroke volume 60 mL. Cardiac output? Ans: 80 x 60 = 4800 mL/min (4.8 L/min).

Q4. Which valves close to produce the second sound (dub)? Ans: Aortic and pulmonary (semilunar) valves.

3D heart model: BodyParts3D, (c) The Database Center for Life Science, licensed under CC BY-SA 2.1 Japan.